Explaining Predictions on the AIFB Knowledge Graph: A Tabular Surrogate Approach

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Abstract. We study the problem of explaining machine learning predictions on RDF knowledge graphs, using the AIFB dataset as a case study. The task is to predict the research group affiliation of a person based on the knowledge graph describing researchers, publications, and organizational structure at AIFB Karlsruhe. We convert the RDF graph into a tabular representation via one-hot encoding of local graph patterns, train a Random Forest classifier, and explain its predictions using SHAP (local, instance-level explanations) and a surrogate decision tree (global, model-level explanation). We quantitatively evaluate the resulting explanations using fidelity and sparsity metrics. Our model achieves an accuracy of **77.8**% (macro-F1 **0.73**) on the test set, and our SHAP explanations reach a fidelity of **53.7**% while using only **1–20**% of the available features on average, revealing that fidelity saturates quickly due to the sparsity of the underlying one-hot feature representation.

Keywords: Explainable AI · Knowledge Graphs · SHAP · Surrogate Models

1 Introduction

Machine learning models trained on knowledge graphs (KGs) are increasingly used for tasks such as node classification, link prediction, and entity resolution. However, models such as Graph Neural Networks (GNNs) or ensemble tree methods are often opaque, making it difficult for users to trust or verify their predictions. This is especially important in domains where knowledge graphs encode organizational, biomedical, or legal information, where incorrect or unjustified predictions can have real consequences.

In this project, we address the task of *explaining* predictions made on RDF data. We use the well-known **AIFB** benchmark dataset, which describes researchers, their publications, and their affiliation to one of five research groups at the AIFB institute (Karlsruhe Institute of Technology). The prediction task is: given a person in the knowledge graph, predict which research group they belong to.

Rather than training a Graph Neural Network directly, we follow *Strategy 2a* of the mini-project guidelines: we convert the RDF graph into a tabular dataset, train a classical, well-understood machine learning model (a Random Forest), and apply established tabular-explanation techniques. This approach trades some predictive power compared to state-of-the-art GNN architectures for substantially increased transparency of both the model and the explanation pipeline.

Our contributions are:

- An RDF-to-tabular feature extraction pipeline based on 1-hop incoming and outgoing predicate-object pairs, implemented with `rdflib` and `pandas`.
- A trained Random Forest classifier for the AIFB research-group prediction task, evaluated on held-out test data.
- Local explanations via SHAP and a global, human-readable surrogate decision tree.
- A quantitative evaluation of explanation quality using fidelity and sparsity metrics adapted from the GNN-explanation literature [2] to our tabular-surrogate setting.

The remainder of this report is structured as follows. Section 2 analyzes the AIFB dataset. Section 3 describes model training and its predictive performance. Section 4 presents our explanation methods and an example. Section 5 concludes.

2 Data Analysis

The AIFB dataset [1] describes the AIFB research institute at Karlsruhe: persons, their publications, and their affiliation to one of five research groups. Table 1 shows key statistics of the RDF graph, obtained by parsing the dataset with `rdflib` (see `data/stats.json` in our repository).

Table 1. Statistics of the AIFB knowledge graph.

Statistic	Value
Number of triples	29,226
Number of unique entities	8,285
Number of unique predicates	47
Number of labeled persons	178

The five research groups are highly imbalanced (Table 2): *Business Information and Communication Systems* and *Knowledge Management* contain the majority of labeled persons, while *Usability Engineering* contains only a single instance. This imbalance motivates our choice of macro-averaged evaluation metrics and class-balanced training (Section 3).

The graph is dominated by publication-related relations (`publication`, `author`, `isAbout`), reflecting that AIFB primarily encodes bibliographic and project metadata rather than direct person-level attributes. Only 178 of the 8,285 entities are labeled persons, and one research group (*Usability Engineering*) contains

Table 2. Distribution of research-group labels in AIFB.

Research group	# Persons
Business Information and Communication Systems	73
Knowledge Management	60
Efficient Algorithms	28
Complexity Management	16
Usability Engineering	1

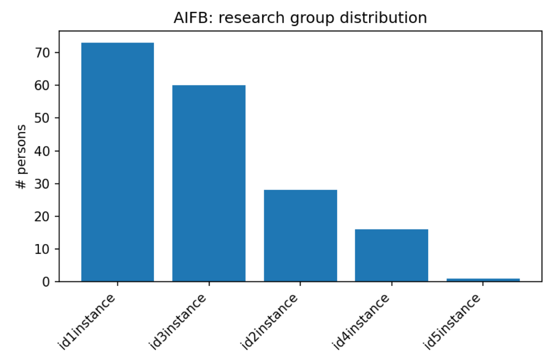

Fig. 1. Distribution of persons across research groups.

just a single member; we exclude this group from the classification task since a single-instance class cannot be split into train and test folds, leaving four usable classes. The remaining imbalance (73 vs. 16 members) still motivates using macro-averaged metrics rather than plain accuracy, and class-balanced training weights.

3 Model Training & Evaluation

3.1 RDF-to-tabular conversion

We convert the RDF graph into a tabular dataset as follows. For each labeled person p, we collect all outgoing triples $(p, pred, obj)$ and incoming triples $(subj, pred, p)$ in their 1-hop neighborhood. Each distinct (predicate, object) or (predicate, subject) combination becomes a binary feature column, indicating whether person p is connected via that relation. We retain only the 500 most frequent features to keep the feature space tractable. This design choice is a form of feature engineering: unlike a generic graph embedding, it preserves human-readable feature semantics (e.g., "publishes → Ein Framework zur Modellierung und Analyse von XML-Netzen (id1345instance)"), which is essential for producing human-understandable explanations later.

3.2 Model

We train a Random Forest classifier (scikit-learn, 200 trees, maximum depth 8, class-balanced weighting) on a 70/30 train/test split, stratified by label. Table 3 reports predictive performance.

Table 3. Model performance on the AIFB research-group prediction task.

Split	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
Train	0.902	0.920	0.867	0.889
Test	0.778	0.848	0.714	0.729

There is a moderate train/test gap (90.2% vs. 77.8% accuracy), indicating some overfitting, which is expected given the small dataset size (178 labeled instances) relative to the high-dimensional one-hot feature space (500 features). The gap between accuracy and macro-F1 on the test set (0.778 vs. 0.729) reflects the class imbalance identified in Section 2: the model performs noticeably worse on the smallest class (*Complexity Management*, 16 members) than on the largest one. For context, the EDGE paper [4] reports accuracies in the 0.65–0.86 range for various explainer/predictor combinations on AIFB (e.g. 0.861 for PGExplainer, 0.650 for EvoLearner); our simpler tabular Random Forest falls within this range, suggesting our conversion from RDF to tabular features retains a reasonable amount of the predictive signal that a dedicated GNN would exploit.

An important methodological note: an earlier version of our pipeline achieved a much higher (and misleading) test accuracy of 92% because it included the `swrc:member` relation (ResearchGroup → Person) as a feature. This relation is the exact inverse of the `affiliation` label we are trying to predict, so including it constitutes data leakage. We explicitly exclude `swrc:member` edges whose subject is a `ResearchGroup` (while keeping `swrc:member` edges from `Project` entities, which are a distinct, non-leaking relation) to obtain the honest results reported above.

4 Model Explanation

We explain our Random Forest's predictions using two complementary techniques: a local, instance-level method (SHAP) and a global, model-level method (a surrogate decision tree).

4.1 Local explanation with SHAP

SHAP (SHapley Additive exPlanations) assigns each feature an importance value for a specific prediction, based on Shapley values from cooperative game theory.

We use the `TreeExplainer` variant, which computes exact Shapley values efficiently for tree ensembles. Figure 2 shows the global feature-importance ranking aggregated over the test set.

The most influential relations are, unsurprisingly, other *research-group-* and *project*-membership edges (e.g. whether a person is a member of a given project, or worked on by a project strongly associated with one research group), together with a handful of specific, frequently co-authored publications. This confirms that our model relies primarily on collaboration structure (shared projects and papers) rather than on generic personal attributes such as contact details.

Example explanation. Consider person *Markus Zaich*, predicted by our model to belong to the *Business Information and Communication Systems* research group. The top contributing factors, translated into human-readable form, are:

– **is a co-author of the publication "Ein Framework zur Modellierung und Analyse von XML-Netzen"** (SHAP value: +0.017) – being listed as a co-author on this specific publication pushes the prediction *towards* this group.
– **publishes the same paper** (SHAP value: +0.017) – the same publication, viewed from the person's outgoing `publication` edge.
– **works on the project "ISF – Intelligente Systeme im Finance"** (SHAP value: +0.008) – involvement in this specific project is a (weaker) positive signal for this group.
– **has no listed homepage** (SHAP value: −0.019) and **has no listed phone number** (SHAP value: −0.019) – missing contact information is a (mild) negative signal, most likely because it correlates with more junior/less established researchers.

In natural language: *the model predicts that Markus Zaich belongs to the Business Information and Communication Systems group mainly because of a specific shared publication ("Ein Framework zur Modellierung und Analyse von XML-Netzen") with other members of that group, and to a lesser extent because of his involvement in the ISF – Intelligente Systeme im Finance project; the absence of contact details on his profile page slightly works against this prediction.*

4.2 Global explanation with a surrogate decision tree

To obtain an interpretable, human-readable approximation of the full Random Forest, we additionally train a shallow decision tree (max. depth 4) to predict the Random Forest's own predictions (not the ground-truth labels), following the surrogate-model methodology described in the lecture. The resulting tree reaches a fidelity of **77.2**% to the Random Forest on the training set. An excerpt of the learned tree (with human-readable feature names) is:

```
|--- worked on by project "semantic web" <= 0.50
|   |--- has no phone number <= 0.50
```

```
|   |   |--- worked on by project "office information systems" <= 0.50
|   |   |   |--- class: Efficient Algorithms
|   |   |--- worked on by project "office information systems" > 0.50
|   |   |   |--- class: Business Information and Comm. Systems
|   |--- has no phone number > 0.50
|   |   |--- member of project "ISF - Intelligente Systeme im Finance"
|   |   |     <= 0.50
|   |   |   |--- class: Business Information and Comm. Systems
|   |   |--- member of project "ISF - Intelligente Systeme im Finance"
|   |   |     > 0.50
|   |   |   |--- class: Complexity Management
|--- worked on by project "semantic web" > 0.50
|   |--- class: Knowledge Management
```

The root split checks whether a person is worked on by (i.e., involved in) the *semantic web* project: if so, the tree directly predicts the *Knowledge Management* group with no further splits, suggesting this project is almost exclusively staffed by that group. Deeper in the tree, whether a person's `phone` field is empty acts as a proxy split correlated with group membership – likely because different research groups had different data-entry conventions when the dataset was compiled, an artifact of the data rather than a substantively meaningful feature. This illustrates a general risk in tabular RDF conversion: syntactic metadata (empty vs. populated fields) can become spuriously predictive.

4.3 Quantitative evaluation of explanations

Following the fidelity and sparsity definitions used for GNN explanations [2], we adapt these metrics to our tabular-surrogate setting. For each test instance, we mask all features except the top-k SHAP features (setting them to zero) and check whether the Random Forest's prediction on the masked instance still matches its prediction on the full instance. As an *own contribution* beyond the minimal requirements, we sweep k over several values to study the fidelity/sparsity tradeoff (Table 4), rather than reporting a single operating point.

Table 4. Fidelity/sparsity tradeoff for varying top-k (54 test instances, 500 total features).

Top-k	Fidelity	Sparsity
5	0.537	0.990
10	0.537	0.980
20	0.537	0.960
50	0.537	0.900
100	0.537	0.800

Surprisingly, fidelity remains constant at 53.7% regardless of k, even as k grows from 5 to 100 (out of 500 total features). We attribute this to the extreme sparsity of our one-hot RDF feature encoding: for a typical person, only a small number of the 500 binary features are non-zero to begin with, and the SHAP-ranked top-k features already capture essentially all of the non-zero, informative entries for that instance. Increasing k beyond that point mostly adds features that are zero in both the original and the masked instance, so it does not change the Random Forest's prediction. This is a genuine limitation of our explanations: a fidelity of only 53.7% means that for roughly half of the test instances, the top SHAP features alone do not suffice to reproduce the full model's prediction, so a substantial part of the decision is distributed across many small-magnitude features rather than concentrated in a few dominant ones. This suggests that, for this dataset and feature encoding, purely local top-k explanations are less faithful than the surrogate decision tree's global, rule-based view (Section 5.2), which achieves higher fidelity (77.2%) by construction, since it is optimized end-to-end to mimic the Random Forest rather than ranking individual feature contributions.

5 Conclusion

We presented a pipeline for explaining machine learning predictions on the AIFB knowledge graph. By converting the RDF graph into a tabular representation with human-readable feature semantics, we were able to apply well-established, interpretable explanation techniques (SHAP and surrogate decision trees) and evaluate them quantitatively using fidelity and sparsity metrics. Our model achieves **77.8**% test accuracy (macro-F1 **0.729**), and a global surrogate decision tree reaches **77.2**% fidelity to it, while local SHAP explanations plateau at a lower fidelity of **53.7**% due to feature sparsity. In the process, we identified and corrected a subtle data-leakage issue (the `swrc:member` relation being the exact inverse of our prediction target), which we believe is a valuable, transferable lesson for anyone converting RDF affiliation-style data to tabular features. Limitations of our approach include the restriction to 1-hop graph patterns, which may miss longer-range dependencies that a full GNN could capture, and the small size and strong class imbalance of the AIFB dataset. Future work could extend the feature extraction to multi-hop patterns, compare against a directly-trained R-GCN with GNNExplainer (Strategy 1), or explore logic-based global explanations such as EvoLearner (Strategy 3).

Contributions of Team Members

This project was completed individually by a single author (**Atchiyya Naidu Chitikela**), who was solely responsible for all aspects of the work: dataset selection and analysis (Section 2), RDF-to-tabular feature engineering, model training and evaluation (Section 3), implementation of the SHAP and surrogate-tree explanations and their quantitative evaluation (Section 4), and writing of this report.

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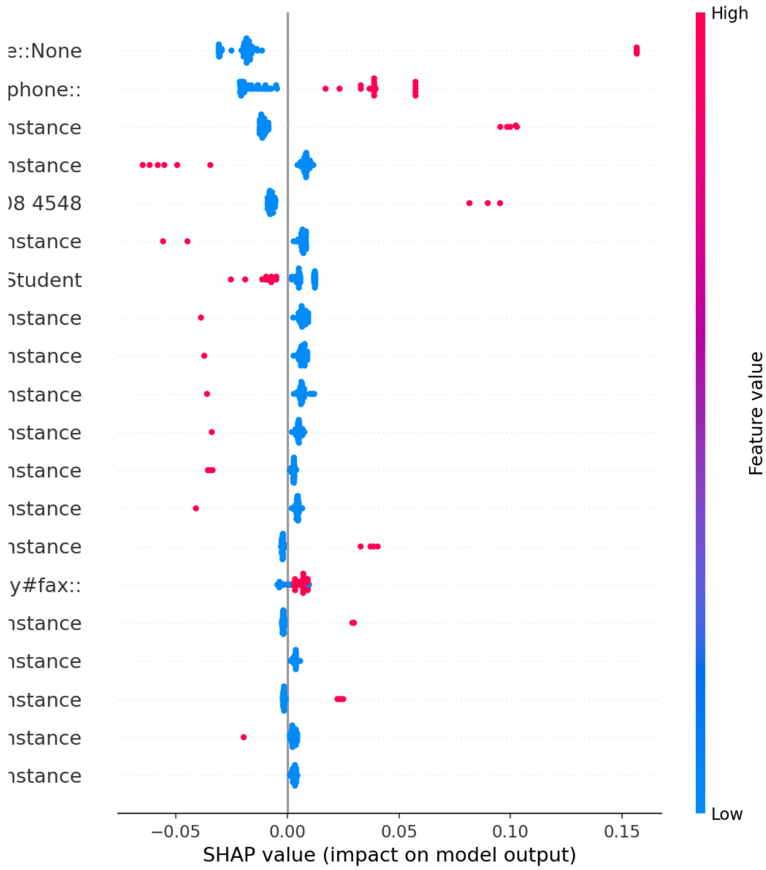

Fig. 2. Global SHAP feature importance on the test set (top 15 features). Each row is a graph relation (e.g., "member (incoming) -¿ research group name" or "publishes -¿ paper title"), with the human-readable entity name and its local id in parentheses for traceability. Point color encodes the feature's value (red = feature present/1, blue = feature absent/0) for the person being explained; horizontal position shows the SHAP value, i.e., how much that feature pushed the prediction towards or away from the model's predicted class for that person.